DELHI PUBLIC SCHOOL JANKIPURAM, LUCKNOW

SESSION 2024-2025: PERIODIC ASSESSMENT-I

SUBJECT: SCIENCE

SET–I

TIME: $1\frac{1}{2}$ Hrs. **M.M.: 40**

Date:_____ Name : _____ Class/Sec IX_____

General Instructions:

Read the following instructions carefully and follow them:

- **The question paper comprises of five sections; A, B, C, D and E. You have to attempt all the sections.**
- **All questions are compulsory.**
- **Q1 to Q9 carry one mark each.**
- **Q10 to Q12 are very short type questions and carry two marks each.**
- **Q13 and Q14 are short type questions and carry three marks each.**
- **Q15 to Q17 are long type questions and carry five marks each.**
- **Q18 is diagram-based study and carry four marks.**

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SECTION A

Select the most appropriate answer. (1x6=6)

1. A car accelerates uniformly from 18 km/h to 36 km/h in 5 minutes. The acceleration is:
(a) $5ms^{-2}$
(b) $1 km\ s^{-2}$
(c) $216\ ms^{-2}$
(d) $216\ km\ h^{-2}$

2. A Motorcycle is being driven at a speed of 20 m/ s when brakes are applied and it is brought to rest in 5 seconds. What is the deceleration produced?
(a) $4m/s^2$
(b) $-4m/s^2$
(c) $0.25\ m/s^2$
(d) $-0.25\ m/s^2$

3. Find out the correct sentence
(a) Enzymes packed in Lysosomes are made through RER (rough endoplasmic reticulum)
(b) Rough endoplasmic reticulum and smooth endoplasmic reticulum produce lipid and protein respectively
(c) Endoplasmic reticulum is related with the destruction of plasma membrane
(d) Nucleoid is present inside the nucleoplasm of eukaryotic nucleus

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4. When some Rheo leaves are boiled in water for few minutes and the killed cells are put in a strong salt solution after mounting. The cells will:
(a) swell up
(b) shrink down
(c) burst
(d) remain unchanged

5. Which condition out of the following will increase the evaporation of water?
(a) Increase in temperature of water
(b) Decrease in temperature of water
(c) Less exposed surface area of water
(d) Adding common salt to water

6. When we put some crystals of potassium permanganate in a beaker containing water, we observe that after sometime whole water has turned pink. This is due to:
(a)Boiling
(b)Melting of potassium permanganate crystals
(c)Sublimation of crystals
(d)Diffusion

Direction (Q7 to 9): In the following questions, there are two statements marked as Assertion and Reason. Mark your answer as per the codes provided below: (1x3=3)
(a)Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.
(b)Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(c) Assertion is true but Reason is false.
(d) Assertion is false but Reason is true.

7. Assertion: A fully turgid cell is called deplasmolysed.
Reason: Deplasmolysis involves exosmosis of water.

8. Assertion: The speedometer of a car measures the instantaneous speed of the car.
Reason: Average speed is equal to the total distance covered by an object divided by the total time taken.

9. Assertion: When a liquid melts, its temperature remains the same.
Reason: The heat gets used up in changing the state by overcoming the forces of attraction between the particles.

SECTION B

10. A car is moving on a straight road with uniform acceleration. The following table gives the speed of the car at various instants of time:

Time(t)	0	10	20	30	40	50
Speed(m/s)	5	10	15	20	25	30

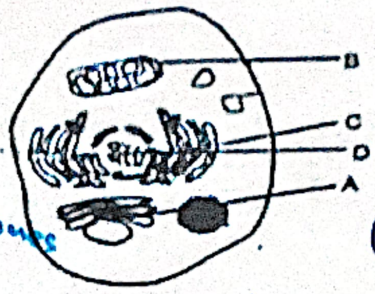
(a)Draw the speed time graph choosing a convenient scale.
(b) Determine from it:
(i) The acceleration of the car.

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(ii) The distance travelled by the car in 50 sec. (2)

11. Label the figure and answer the questions: (2)

(a) A – It is the packaging organelle Lysosome
(b) B – Provides energy Mitochondria
(c) C – helps in the transport of material ER
(d) D – Carries the information. Nucleas / chromosomes



Animal Cell

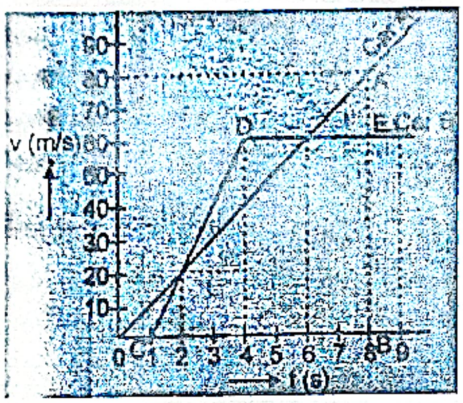
12. Convert the following temperature:
(a) 45°C to Kelvin 0 + 273
(b) 298 K to Celsius

273
273 + 45 = 318 (2)
273 + 8 = 391

SECTION C

13. The v-t graph of cars A and B which start from the same place and move along straight road in the same direction, is shown. Calculate

(i) The acceleration of car A between zero and 8s. 10m/s²
(ii) The points of the time at which both the cars have the same velocity, 2nd second and 6th
(iii) Which of the two cars is ahead after 8s and by how much? Car A by 30m/s (3)



14. (a) Why are we able to sip hot tea or milk faster from a saucer rather than a cup?
(b) Write any two characteristics of the matter. (3)

SECTION D

15. Any change in the velocity of an object results in an acceleration. Increasing speed (what people usually mean when they say acceleration), decreasing speed, or changing direction. Yes, that is right, a change in the direction of motion results in an acceleration even if the moving object neither speed up nor slowed down. That is because acceleration depends on the change in velocity and velocity is a Vector quantity. Thus, a falling apple accelerates; a car stopping at a traffic light accelerates, and the moon in orbit around the earth accelerates. Acceleration occurs anytime an object's speed increases or decreases or it changes direction.

i. Acceleration can be defined as rate of change of:

a. displacement b. velocity c. momentum d. inertia

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ii. Acceleration is a vector quantity which indicates that its value:

a. can be positive, negative or zero
b. is always positive
c. is always negative
d. is zero

iii. A falling apple accelerates:

a. negatively and uniformly
b. positively and uniformly
c. positively and non-uniformly
d. negatively and non-uniformly

iv. Moon orbiting around the earth because of:

a. zero acceleration
b. centripetal acceleration
c. neutral acceleration
d. centrifugal acceleration

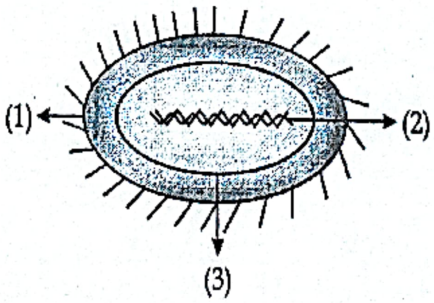
v. Define average speed. Write its formula and SI unit. (1x5=5)

16. Name the cell organelle for the following: (5)
(a) Present only in plant cell, provides strength and rigidity to the cell.
(b) It is the site for lipids synthesis and helps in detoxification of drugs.
(c) The inner membrane is folded to form cristae, it has its own DNA and proteins.
(d) It helps in the formation of lysosomes.
(e) It imparts colour to the fruit and flowers.

17. (a) Define the following (i) Latent heat of fusion (ii) Diffusion (iii) Vaporisation.
(b) A sample of water under study was found to boil at 102 °C at normal temperature and pressure. Is the water pure? Will this water freeze at 0 °C? Give reason for your answer. (3+2)

SECTION E

18. Study the given diagram of bacterial cell and answer any four questions from (i) to (v). 4



(i) Label the parts marked 1, 2, and 3.
(a) 1- Nucleus, 2-Cell wall, 3- Plasma membrane. (b) 1-Cell wall, 2- Nucleoid, 3- Plasma membrane.
(c) 1- Cytoplasm ,2- Nucleus, 3- Cell wall. (d) 1- Nucleus, 2-Cytoplasm, 3- Cell wall.

(ii) Which structure present in the region 2 of a living cell bear genes?
(a) Chromosome (b) Plasma membrane (c) Cytoplasm (d) Cell wall

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(iii) How nuclear region of a bacterial cell and nuclear region of an animal cell is different from each other?
(a) Nuclear region of bacterial cell is well defined but lacks any covering while nuclear region of an animal cell is poorly defined and membrane bound.
(b) Nuclear region of bacterial cell is poorly defined and has a covering while nuclear region of an animal cell is poorly defined and membrane bound.
(c) Nuclear region of bacterial cell is poorly defined and lacks any covering while nuclear region of an animal cell is well defined and membrane bound.
(d) Nuclear region of bacterial cell is well defined and lack any covering while nuclear region of an animal cell is well defined and membrane bound.

(iv) Which of the following statement is correct?
(a) A single cell cannot perform all the life processes independently.
(b) A single cell can perform all the life processes independently.
(c) A single bacterial cell always needs another cell to carry out the life processes.
(d) A bacterium is a multicellular organism.

(v) An example of single celled organism is:
(a) Bird (b) Fish (c) Snake (d) Amoeba

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